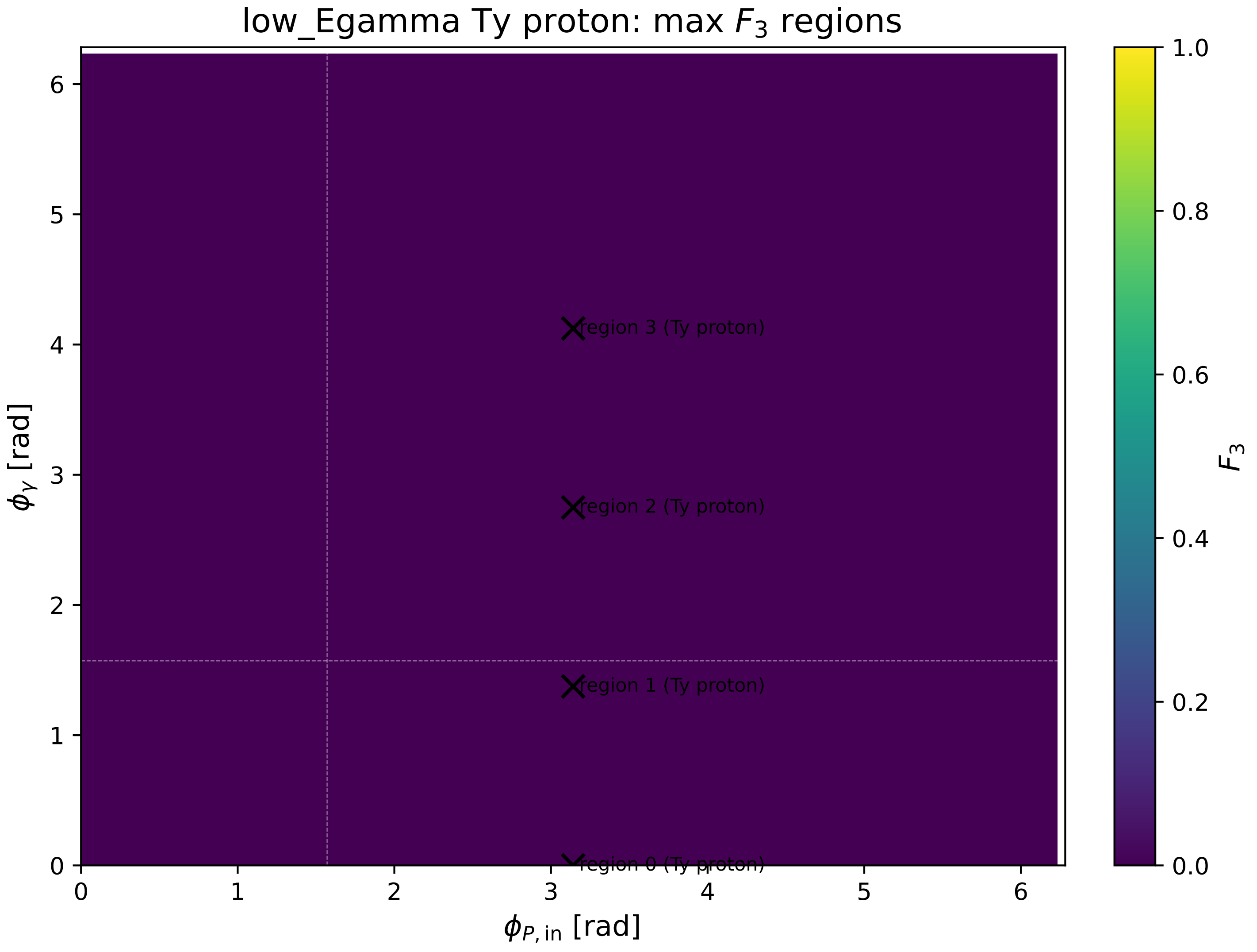
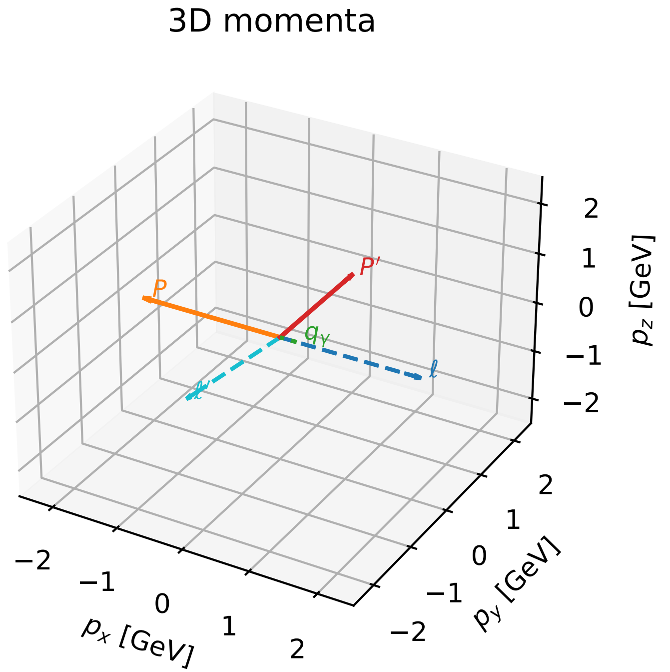


low_Egamma Ty proton: max F_3 regions
region 3 (Ty proton)
region 2 (Ty proton)
region 1 (Ty proton)
region 0 (Ty proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
F_3
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

Ty proton characteristic max F_3 configuration

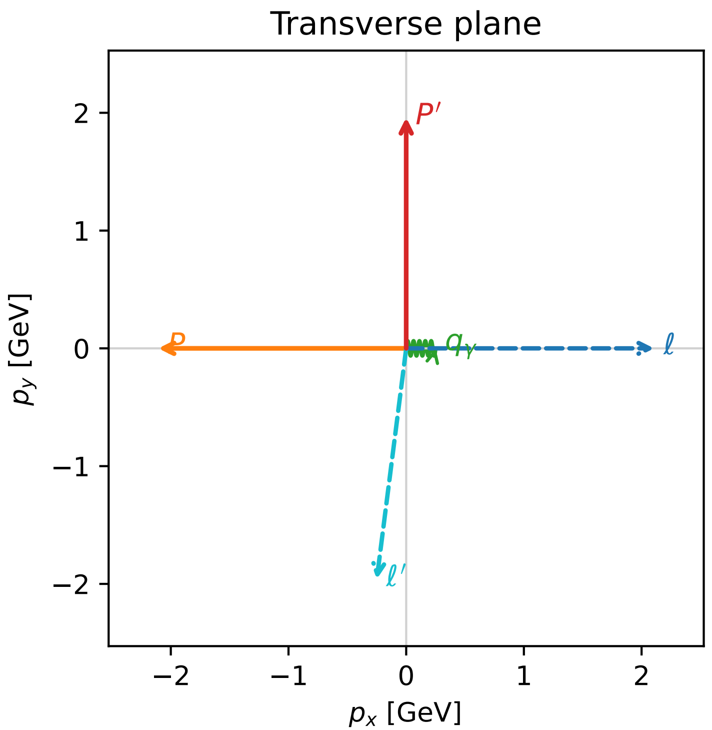


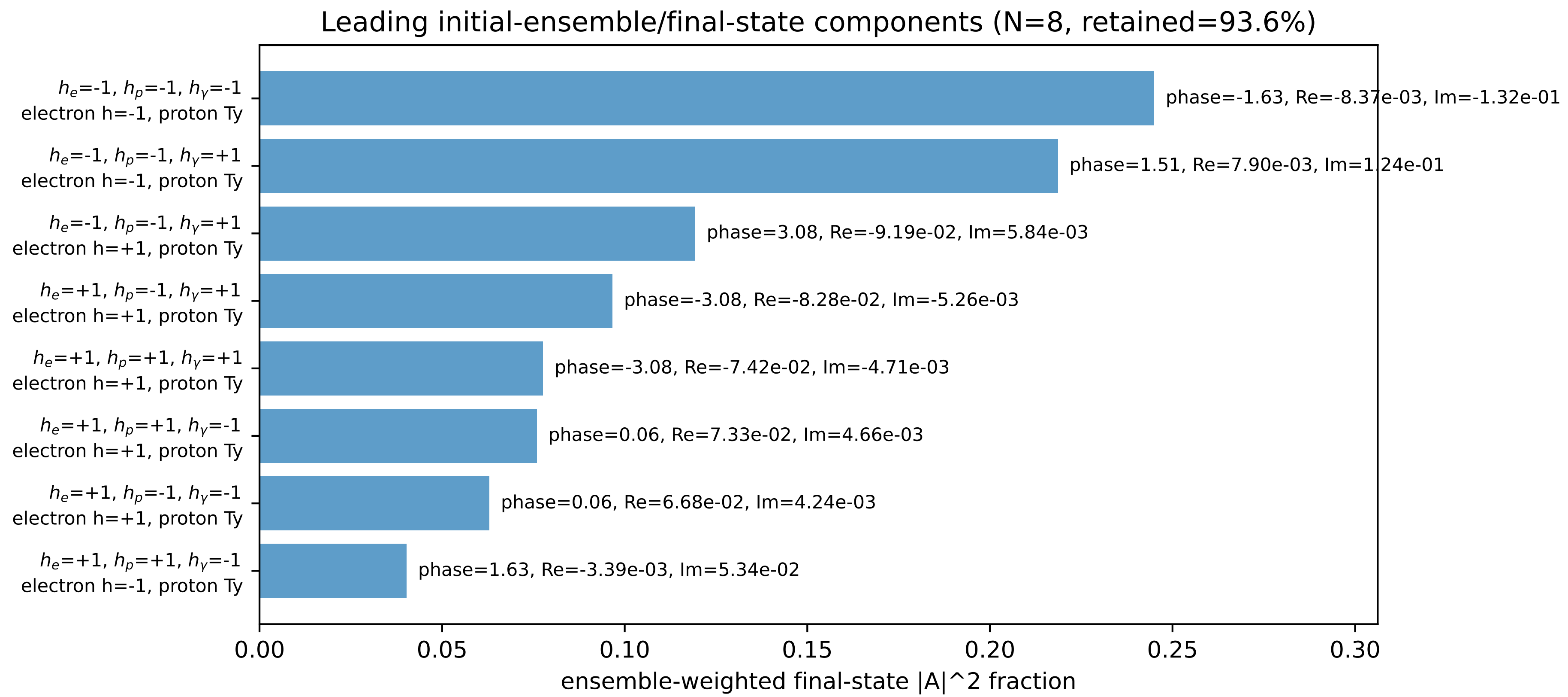

```
f3_low_egamma_ty_proton_region_0  F_3=0
region: low_Egamma_Ty_proton_region_0
outgoing-state purity: 0.515022
kinematic point: high_s_high_theta_in_low_Egamma
```

incoming state: $\frac{1}{2}\sum_{h_\ell}\rho(h_\ell, T_{y,p})$

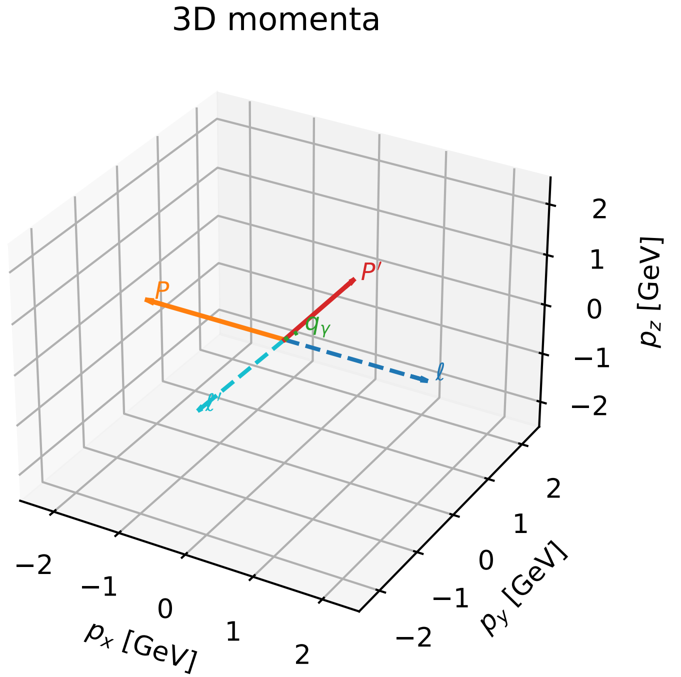
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=1.96062
$\theta_{\rm in}$=1.57079, ϕ_ℓ=0, ϕ_P=3.14159
E_γ=0.25, ϕ_γ=0
Q^2=9.38549, x_B=0.896228, t=-8.26559, W^2=1.96657, y=0.533317
$\theta(\ell', \gamma)$=97.2666 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= 2.1069 p_y= -0.0000 p_z= -0.0000
P E= 2.3063 p_x= -2.1069 p_y= 0.0000 p_z= 0.0000
ℓ' E= 2.2151 p_x= -0.2500 p_y= -1.9606 p_z= 0.0000
P' E= 2.1734 p_x= 0.0000 p_y= 1.9606 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.2500 p_y= 0.0000 p_z= 0.0000





Ty proton characteristic max F_3 configuration

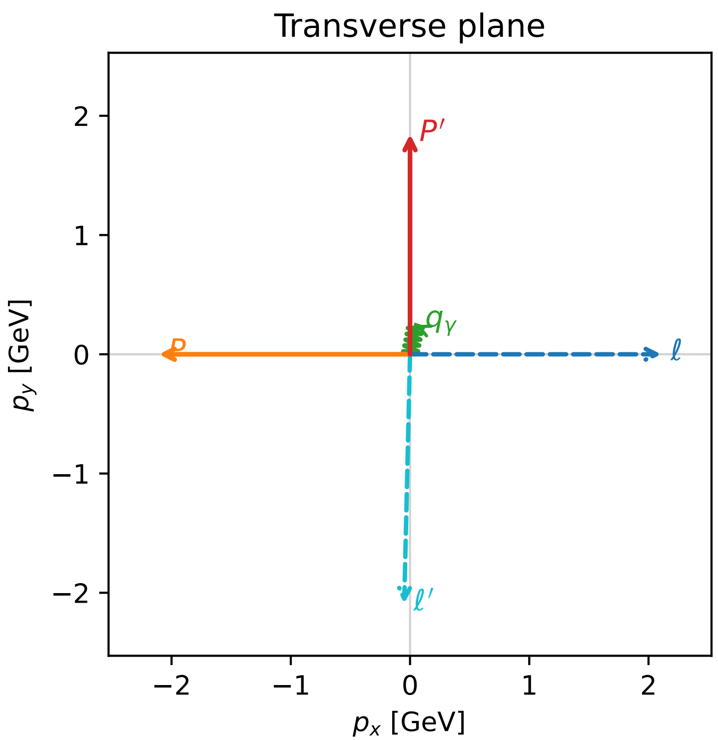


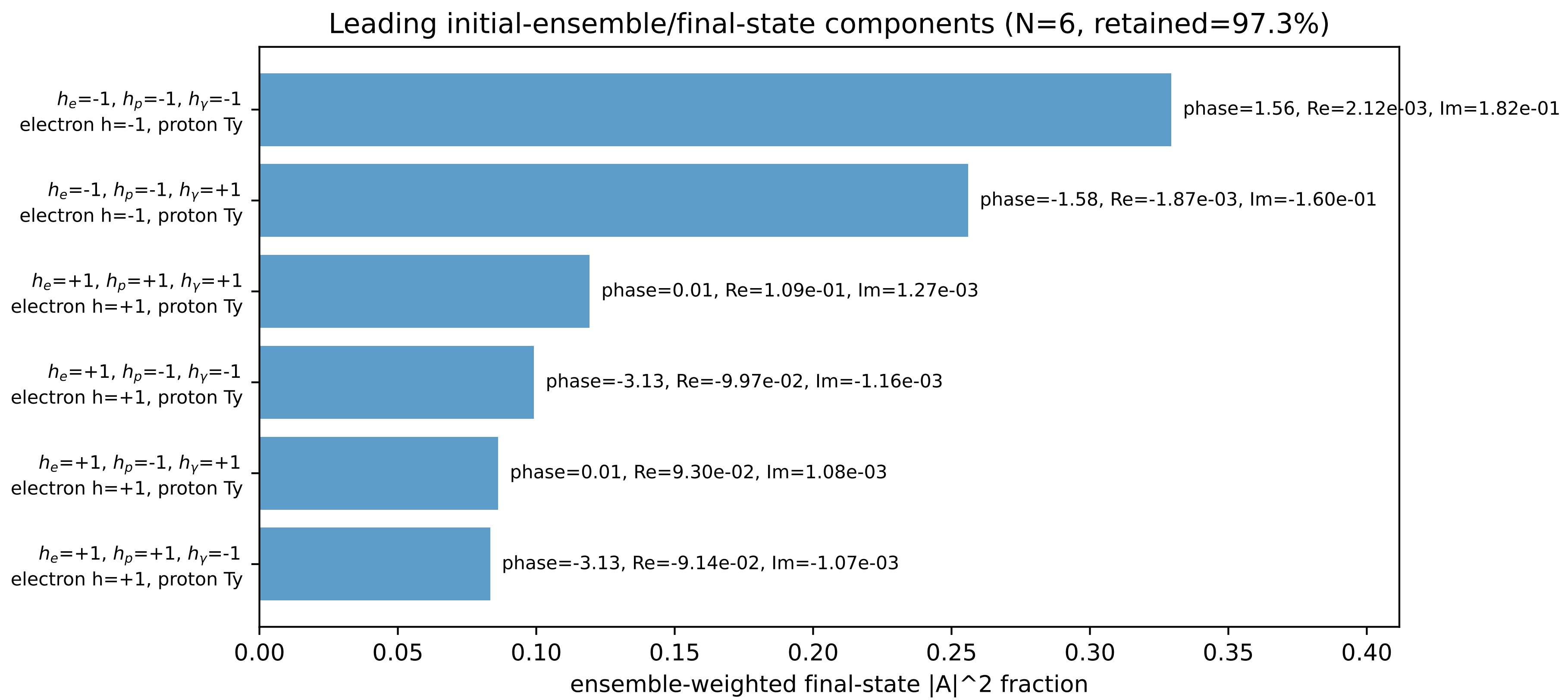

```
f3_low_egamma_ty_proton_region_1  F_3=0
region: low_Egamma_Ty_proton_region_1
outgoing-state purity: 0.521787
kinematic point: high_s_high_theta_in_low_Egamma
```

incoming state: $\frac{1}{2}\sum_{h_\ell}\rho(h_\ell, T_{y,p})$

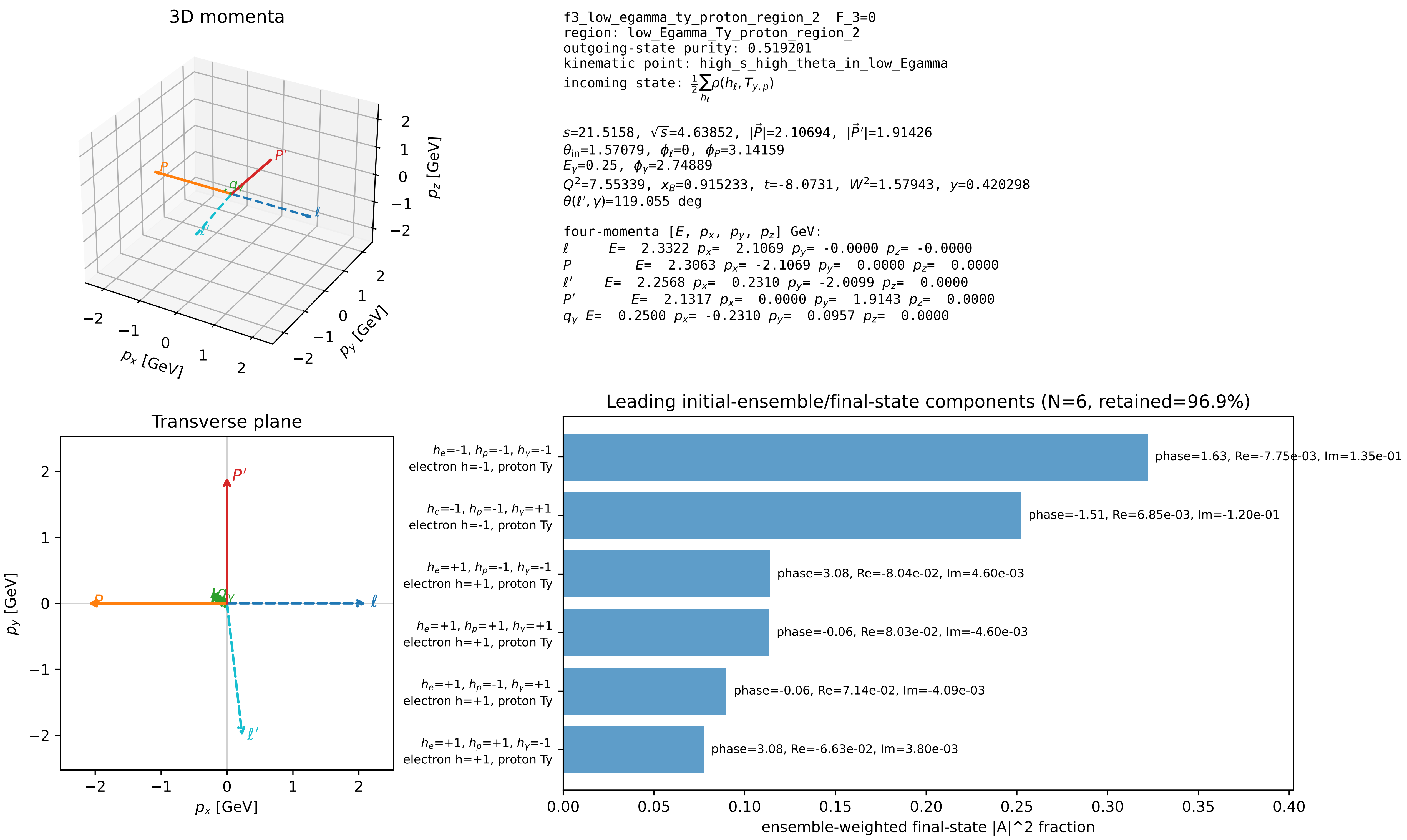
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=1.84565
$\theta_{\rm in}$=1.57079, ϕ_ℓ=0, ϕ_P=3.14159
E_γ=0.25, ϕ_γ=1.37445
Q^2=9.01854, x_B=0.985782, t=-7.78993, W^2=1.00992, y=0.46591
$\theta(\ell', \gamma)$=170.086 deg

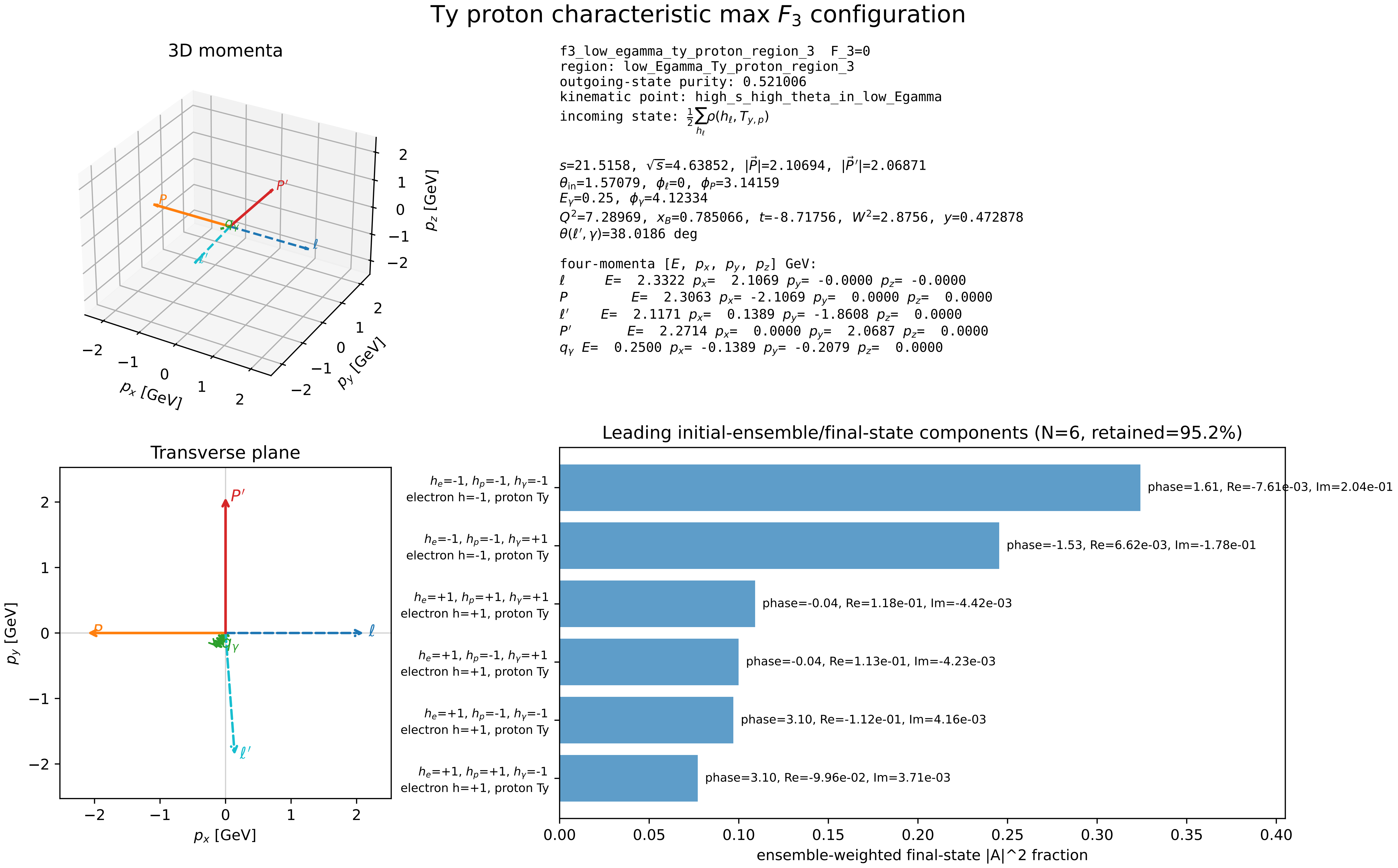
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= 2.1069 p_y= -0.0000 p_z= -0.0000
P E= 2.3063 p_x= -2.1069 p_y= 0.0000 p_z= 0.0000
ℓ' E= 2.3182 p_x= -0.0488 p_y= -2.0908 p_z= 0.0000
P' E= 2.0703 p_x= 0.0000 p_y= 1.8456 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.0488 p_y= 0.2452 p_z= 0.0000





Ty proton characteristic max F_3 configuration




Ty proton characteristic max F_3 configuration
3D momenta
f3_low_egamma_ty_proton_region_3 F_3=0
region: low_Egamma_Ty_proton_region_3
outgoing-state purity: 0.521006
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_\ell}\rho(h_\ell, T_{y,p})$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=2.06871
θ_{in}=1.57079, ϕ_ℓ=0, ϕ_P=3.14159
E_γ=0.25, ϕ_γ=4.12334
Q^2=7.28969, x_B=0.785066, t=-8.71756, W^2=2.8756, y=0.472878
$\theta(\ell', \gamma)$=38.0186 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= 2.1069 p_y= -0.0000 p_z= -0.0000
P E= 2.3063 p_x= -2.1069 p_y= 0.0000 p_z= 0.0000
ℓ' E= 2.1171 p_x= 0.1389 p_y= -1.8608 p_z= 0.0000
P' E= 2.2714 p_x= 0.0000 p_y= 2.0687 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.1389 p_y= -0.2079 p_z= 0.0000
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
Leading initial-ensemble/final-state components (N=6, retained=95.2%)
h_e=-1, h_p=-1, h_γ=-1
electron h=-1, proton Ty
phase=1.61, Re=-7.61e-03, Im=2.04e-01
h_e=-1, h_p=-1, h_γ=+1
electron h=-1, proton Ty
phase=-1.53, Re=6.62e-03, Im=-1.78e-01
h_e=+1, h_p=+1, h_γ=+1
electron h=+1, proton Ty
phase=-0.04, Re=1.18e-01, Im=-4.42e-03
h_e=+1, h_p=-1, h_γ=+1
electron h=+1, proton Ty
phase=-0.04, Re=1.13e-01, Im=-4.23e-03
h_e=+1, h_p=-1, h_γ=-1
electron h=+1, proton Ty
phase=3.10, Re=-1.12e-01, Im=4.16e-03
h_e=+1, h_p=+1, h_γ=-1
electron h=+1, proton Ty
phase=3.10, Re=-9.96e-02, Im=3.71e-03
ensemble-weighted final-state |A|^2 fraction